

Discussion with Dr Ersoy

Jason Duran Dutton | Research discussion brief | 23 September 2026

What I want to investigate

My explicit goal is to derive physical matter from a mathematically consistent field framework motivated by a primordial unity hypothesis, and to determine whether its excitations can generate a geometry universally experienced by matter. I want to pursue this as a falsifiable research direction. I do not want my earlier SEFI, DEFI or GWFM work treated as established physics, and I am seeking critical guidance on which parts are mathematically sound, physically meaningful and worth developing.

What I can presently show

Starting from a canonical rotating complex scalar field on an assumed flat spacetime, we obtain an effective propagation metric for its low-energy phase sector. A traveling-profile transformation leads to a Gross-Pitaevskii vortex-ring equation. Numerical moving rings exist on the tested finite grids.

A provisional second complex field supports charge localized inside a vortex core. We solved the two fields together to obtain a moving charged-loop candidate. In one nonlinear axisymmetric disturbance test, the loop retains winding one through time 100. Time-step refinement agrees closely, and the conserved carrier charge is monitored. Two additional linear azimuthal disturbance samples show modest finite-time amplification.

The most important qualification is analytical: with the original carrier self-coupling, the frequency-constrained potential is nonnegative throughout the localization range. A scaling argument excludes the localized time-harmonic rest state we were seeking. Changing that coupling can remove the obstruction while preserving the bulk state, but the first two revised rest-loop searches did not establish a solution.

What I am not claiming

I have not derived an electron or another observed particle, electric charge, quantum spin or statistics, universal gravity, or a Big Bang evolution. The second field is an added assumption. The numerical loop has finite-domain uncertainty and has not passed a full three-dimensional stability analysis. A classical winding number is not a derivation of quantum spin, and a global conserved charge is not automatically electromagnetism.

Questions I would like to discuss

1. Is the starting canonical field model an appropriate research vehicle for this goal, or does the assumed rotating background already constrain it to an effective-medium interpretation?
2. Is the rest-state scaling obstruction correctly formulated for fixed-charge extrema and the stated boundary conditions? Are there physically justified ways around it that do not add arbitrary structure?
3. Does the revised parameter window justify a constrained continuation study of charged loops, or should the next step be a different class of localized solutions? How should we distinguish numerical failure from evidence of nonexistence?

4. What stability analysis would be sufficient before presenting the loop as a credible classical excitation: spectral analysis, full three-dimensional evolution, constrained energy methods, or a combination?
5. What is the minimum defensible bridge from this classical excitation to physical matter? Which observable, symmetry, quantization condition or coupling should be addressed first?
6. The phase sector has an effective metric, while other degrees of freedom do not yet share it. Is this universality problem the central obstruction to the wider spacetime claim?
7. Could the present work support a narrowly scoped mathematical or computational paper after independent validation? I would value advice on novelty relative to analogue gravity, superconducting strings and vortons, rather than assuming that the broad unification goal is already publishable.

A useful outcome from the meeting

I would like to leave with one prioritized next calculation, an agreed criterion for success or rejection, and guidance on the scope of a defensible manuscript. The accompanying research-direction document contains the equations, parameter choices, numerical results, failed searches and references. The independent research code is kept separate from the existing SEFI-PY engine.

Suggested opening statement: "I am trying to turn a broad field-unity idea into a constrained physics research program. I now have a concrete classical model, a charged moving-loop candidate, and a no-go result for one proposed rest branch. I would like your help deciding which mathematical step would most meaningfully test whether this can progress toward physical matter."